Laura Greenstreet

I am a PhD student at Cornell developing AI and optimization methods with applications in biology, including fishery sciences, ecology, and genomics. Previously, I worked with Dr. Claire Kremen to scale an analysis of functional connectivity to the global level and with Dr. Geoffrey Schiebinger to develop optimization methods for single-cell genomics.

Education

2021 - Present	**Ph.D. Computer Science**, Cornell University Research areas: Optimization, Deep Learning, AI/ML for Science
2020	**B.Sc. Honours Computer Science, Mathematics Minor**, University of British Columbia

Publications

- 2025 F.S. Pacheco, S.A. Heilpern, C. DiLeo, ..., **L. Greenstreet**, et al. *Towards sustainable aquaculture in the Amazon.* Nature Sustainability, 2025. https://doi.org/10.1038/s41893-024-01500-w
- 2025 Li, H., P. Cote, M. Kuoch, J. Ezike, ..., **L. Greenstreet**, et al. *The dynamics of hematopoiesis over the human lifespan.* Nature Methods, 2025. https://doi.org/10.1038/s41592-024-02495-0
- 2025 Massri, A.J., A. Berrio, A. Afanassiev, **L. Greenstreet**, et al. *Single-Cell Transcriptomics Reveals Evolutionary Reconfiguration of Embryonic Cell Fate Specification in the Sea Urchin Heliocidaris erythrogramma.* Genome Biology and Evolution, 2025. https://doi.org/10.1093/gbe/evae258
- 2023 **Greenstreet, L.**, A. Afanassiev, Y. Kijima, M. Heitz, S. Ichiguro, et al. *DNA-GPS: A Theoretical Framework for Optics-Free Spatial Genomics and Synthesis of Current Methods.* Cell Systems, 2023. https://doi.org/10.1016/j.cels.2023.08.005
- 2023 T.M. Nolan, N. Vukasinovic, C.W. Hsu, J. Zhang, $\cdots$, **Laura Greenstreet**, et al. *Brassinosteroid gene regulatory networks at cellular resolution in the Arabidopsis root.* Science, 2023. https://doi.org/10.1126/science.adf4721
- 2023 Mirka, R., **L. Greenstreet**, M. Grimson, C.P. Gomes. *A New Approach to Finding 2 x n Partially Spatially Balanced Latin Rectangles*, CP, 2023.
- 2022 Brennan, A., R. Naidoo, **L. Greenstreet**, Z. Mehrabi, N. Ramankutty, C. Kremen. *Functional Connectivity of the Worlds Protected Areas.* Science, 2022. https://doi.org/10.1126/science.abl8974
- 2022 **Greenstreet, L.**, N.J.A. Harvey, V. Sanches Portella. *Efficient and Optimal Fixed-Time Regret with Two Experts.* ALT, 2022. https://doi.org/10.48550/arXiv.2203.07577
- 2021 Zhang, S., A. Afanassiev, **L. Greenstreet**, T. Matsumoto, G. Schiebinger. *Optimal transport analysis reveals trajectories in steady-state systems.* PLOS Computational Biology, 2021. https://doi.org/10.1371/journal.pcbi.1009466
- 2021 Li, H., J. Ezike, A. Afanassiev, **L. Greenstreet**, et al. *Single Cell Analysis Elucidates the Maturation of Human Stem and Progenitor Cell Function from Fetal through Adult Hematopoiesis.* Blood, 2021. https://doi.org/10.1182/blood-2021-151090
- 2021 Shahan R., C. Hsu, T.M. Nolan, B.J. Cole, I.W. Taylor, **L. Greenstreet**, et al. *A single cell Arabidopsis root atlas reveals developmental trajectories in wild type and cell identity mutants.* Developmental Cell, 2021. https://doi.org/10.1016/j.devcel.2022.01.008

Bioinformatics Geospatial Theory

- 2020 Massri, A.J., **L. Greenstreet**, A. Afanassiev, A. Berrio Escobar, G.M. Wray, G. Schiebinger, D.R. McClay. *Developmental Single-cell transcriptomics in the* Lytechinus variegatus *Sea Urchin Embryo.* Development, 2020. https://doi.org/10.1242/dev.198614

Workshops and Presentations

- 2023 **Greenstreet, L**, F. Pacheco, J. Fan, M. Eichemberger Ummus, et al. *Detecting Aquaculture with Deep Learning in a Low-Data Setting.* AGU Fall Meeting Abstracts.
- 2023 **Greenstreet, L**, J. Fan, F. Siqueira Pacheco, Y. Bai, M. Eichemberger Ummus, et al. *Detecting Aquaculture with Deep Learning in a Low-Data Setting.* SigKDD 2023 Fragile Earth Workshop.
- 2020 **Greenstreet, L**, and E. Lai. *Developing a Data-Driven Electric Vehicle Strategy in Surrey, BC.* SigKDD 2020 Social Impact Session.

Awards

2022-2024	**Graduate Teaching Award x4**, Department of Computer Science, Cornell University
2020	**Undergraduate Summer Research Award**, Natural Sciences and Engineering Research Council of Canada (NSERC)
2019	**Data Science for Social Good Fellowship**, UBC Data Science Institute
2018	**Stanley M Grant Scholarship in Mathematics**, UBC Department of Mathematics

Technical Skills

3+ Years Experience: Python, Git, Linux

1-3 Years Experience: R, Julia, Matlab, Java, SQL

Libraries/Tools: Pytorch, QGIS, Postgres, Gurobi

Experience: deep learning: CNNs, GNNs, VAEs, transformers, contrastive learning, manifold learning, mathematical programming, SQL

Research Experience

05/2024 - 08/2024	**Graduate Research Assistant**, Cornell University
05/2023 - 01/2024 05/2022 - 08/2022	**Graduate Research Assistant**, Computational Sustainability Lab, Department of Computer Science, Cornell University
05/2020 - 08/2021	**Research Assistant**, Schiebinger Lab, Department of Mathematics, University of British Columbia
09/2019 - 09/2020	**Research Assistant**, WoRCS Lab, Institute for Resources, Environment, and Sustainability, University of British Columbia
05/2019 - 08/2019	**Fellow - Data Science for Social Good Program**, University of British Columbia Data Science Institute

Teaching Experience

FA23, SP23, SP24	**Head Teaching Assistant**, Cornell University, Ithaca, NY CS 2700 - Excursions in Computational Sustainability CS 4700/4701 - Foundations/Practicum in Artificial Intelligence
FA22, SP22, SP25	**Teaching Assistant**, Cornell University, Ithaca, NY CS 4820 - Introduction to Analysis of Algorithms CS 4220 - Numerical Analysis: Linear and Nonlinear Problems CS 3220 - Computational Mathematics for Computer Science

Community Involvement

2024	**Organizer**, AI for Science Seminar
2023	**Organizer**, NeurIPS Computational Sustainability Workshop
2023	**Mentor**, BURE Undergraduate Research Program, Cornell
2023-2024	**Assistant Organizer**, AI for Science Program, Cornell
2020-2021	**Mentor**, Data Science for Social Good Program, UBC Data Science Institute